

$\neg(\neg(x \in B))$ is False. By Double-negation Law of Propositions, $x \in B$ is False.
 By Defn. of AND, $x \in A \wedge x \in B$ is False. By Defn. 2, $x \in (A \cap B)$ is False.
 By Defn. of empty set, $(A \cap B) = \emptyset$.
 Let x be an arbitrary element of $(A \cup B) \cap (\overline{A \cup B})$, i.e. $x \in (A \cup B) \cap (\overline{A \cup B})$.
 By Defn. 2, $x \in (A \cup B) \wedge x \in (\overline{A \cup B})$. By Defn. of AND, $x \in (A \cup B)$ and $x \in (\overline{A \cup B})$.
 By Defn. of union, $(x \in A \vee x \in B)$ and $(x \in \overline{A \cup B})$.
 Let $x \in B$. This is consistent with the defn. of OR with $x \in A \vee x \in B$.
 By Defn. of negation, $\neg(x \in B)$ is False. By Defn. of not belongs to, $x \notin B$.
 is False. By Defn. 5, $x \in \overline{B}$ is False. By Defn. of OR, $x \in A$ is True.
 $\therefore$ Defn. 5 gives $x \notin A$. By Defn. of not belongs to, $\neg(x \in A)$ is True. By
 Defn. of negation, $\neg(\neg(x \in A))$ is False. By Double-negation Law of
 Propositions, $x \in A$ is False.
 $\therefore (A \cup B) \cap (\overline{A \cup B}) \subseteq A$, By Defn. 3 from Sec-2.1, $x \in A$. This is a contradiction.
 $\therefore x \in B$ is False. By Defn. of AND, $x \in A \wedge x \in B$ is False. By Defn. 2, $x \in (A \cap B)$
 is False. By Defn. of empty set, $(A \cap B) = \emptyset$.
 Given, $(A \cup B) - (A \cap B) = A \Rightarrow (A \cup B) - \emptyset = A \Rightarrow \boxed{A \cup B = A}$ (Prob 10(a))
 $\therefore A \cup B \subseteq A$ and $A \subseteq A \cup B$.
 Let x be an arbitrary element of B , i.e. $x \in B$. By Defn. of OR,
 $x \in A \vee x \in B$. By Defn. 1, $x \in (A \cup B)$. By Defn. 3 from Sec-2.1, $x \in A$.
 By Defn. of AND, $x \in A \wedge x \in B$. By Defn. 2, $x \in (A \cap B)$. We know that
 $(A \cap B) = \emptyset$. $\therefore x \in B$ is a False assumption. By Defn, $\boxed{B = \emptyset} \quad \square$

4) Determine whether the symmetric difference is associative; that is, if
 A, B and C are sets, does it follow that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$?
 Solution: $A \oplus (B \oplus C) = A \oplus ((B \cup C) - (B \cap C))$ (Prob 41)
 $= A \oplus ((B \cup C) \cap (\overline{B \cap C}))$ (Prob 21(a)) $= A \oplus ((B \cup C) \cap (\overline{B \cap C}))$
 $= (A \cup ((B \cup C) \cap (\overline{B \cap C}))) - (A \cap ((B \cup C) \cap (\overline{B \cap C})))$ (Prob 41) (1st De Morgan's Law)